RIN RAY

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Mathematician and computational scientist · PhD, Northwestern · Universität Münster

PROFILE

I work on what can be learned from the traces systems leave behind. Over more than a decade, my work has crossed computational physics, autonomous robotics, biomedical sensing, and machine learning on biological signals, building models and instruments for working with partial and indirect information. I came to mathematics from the laboratory and never really left; I am now a postdoctoral fellow in homotopy theory and arithmetic geometry, alongside active research in biological systems.

RESEARCH

LIVING THINGS

Hidden states, read from the traces a system emits.

2024– CURRENT — **Sensory Sensitivity as a Unifying Mechanism**; with L. Estinto; synthesis arguing that failure of habituation is a shared upstream mechanism, drawing on connectivity imaging, glutamatergic excitability, microbiome composition and enteric regulation of immunoglobulin secretion. Hypothesis-generating; not experimentally validated.

systems biology · neuroimmunology · sensory processing and habituation · gut microbiome · autoimmunity · chronic pain · glutamatergic excitability · evidence synthesis

2026 CURRENT — **Neostigmine in feline idiopathic megacolon**; owner-reported case study of an acetylcholinesterase inhibitor used off-label to restore colonic motility, documented to a standard usable in veterinary consultation. A treatment that worked and had not been written down.

veterinary pharmacology · enteric motility and cholinergic transmission · acetylcholinesterase inhibition · case documentation of an unpublished treatment

2026 CURRENT — **Memantine as a candidate non-opioid analgesic in central sensitization**; review of the mechanistic rationale, comparative pharmacology against ketamine and dextromethorphan, and the clinical trial record for memantine in chronic pain, with a proposed design for a publicly funded trial. Evidence synthesis only; the pooled analysis of eleven trials is not significant.

pharmacology · NMDA receptor antagonism · central sensitization · opioid tolerance and hyperalgesia · drug repurposing · trial design · health policy

2013–2014 — **Unsupervised classification of rodent ultrasonic vocalizations**; Vium; quasi-real-time clustering of calls from continuously monitored cages, deployed as a non-invasive preclinical readout for compound effects on stress and libido. Run continuously, not on a corpus after the fact, several years before MUPET and DeepSqueak made unsupervised clustering standard. Found a previously uncharacterized female–female vocalization.

computational bioacoustics · unsupervised clustering of ultrasonic vocalizations (USVs) · quasi-real-time audio processing · affective-state readout · preclinical safety pharmacology · 3Rs refinement

2014 — **Decoder convergence for intracortical motor prostheses**; convergence analysis of Kalman-style decoders under signal drift: how fast the filter settles, how far it degrades as glial scarring thins the recorded population, and how much adaptive recalibration recovers. Then optical recording via fluorescent calcium indicators as an alternative to penetrating electrodes.

intracortical brain–computer interfaces · decoder convergence analysis · adaptive filtering under non-stationary signals · foreign body response and chronic signal loss · optical recording

2011–2014 — **Portable detection of gluten and allergenic proteins**; a laser and avalanche-photodiode optical assay, abandoned for insufficient signal-to-noise against protein-rich matrices; then a G12 antibody colorimetric assay. The limit was antibody cost per assay and sampling variance in food, not sensitivity.

point-of-consumption allergen detection · laser absorption spectroscopy · G12 antibody colorimetric immunoassay · assay cost and sampling limits

2013 — **Remote photoplethysmography on a head-mounted display**; HackMIT; cardiac pulse recovered from video without contact. Withdrawn on ethical grounds: the measured party cannot detect or refuse the measurement.

remote photoplethysmography from video · contactless physiological sensing · computer vision · consent and surveillance ethics

BODIES IN MOTION

A body reading the traces of its own movement.

2013–2015 — **Modular robotic retrofits for powered wheelchairs**; four assemblies that bolt onto a chair already fitted to its user: automated pressure redistribution, powered seat elevation, assisted transfer, rough-terrain drive. Requirements elicited from wheelchair users with spinal cord injury. Written as a nonprovisional patent application, deliberately not filed, and released under Creative Commons.

assistive robotics · modular retrofit design · soft robotics · pressure injury prevention · rehabilitation engineering · user-led requirements · gaze-controlled mobility (mentorship)

2012 **Graceful degradation under sensor failure: manipulation on proprioception alone**; George Washington University; with vision and all external feedback disabled, a Willow Garage PR2 learned a shape-matched insertion task via reinforcement learning from commanded joint positions and gripper sensing only. Also two-arm load equalization for objects with off-center mass.

graceful degradation · proprioception-only contact-rich manipulation · blind peg-in-hole insertion · reinforcement learning (policy search) · two-arm load equalization · human–robot interaction

2012 **Monocular visual SLAM with motion planning on a micro aerial vehicle**; George Mason University; PTAM-class SLAM processed off-board, metric scale recovered by fusing sonar altimetry and inertial measurement, coupled to frontier-based planning on a partially built map.

simultaneous localization and mapping · monocular visual SLAM with parallax bootstrapping · loop closure · sonar and inertial sensor fusion · motion planning on a partial map · unmanned aerial vehicles

2011 **Gait-based re-identification for robot person following**; hexapedal platform; target enrolled by proximity and re-identified by gait, not appearance.

robot person following · gait-based re-identification · computer vision · statically stable hexapedal locomotion

MATTER

Behaviour inferred from the traces that structure leaves.

2012–2013 **Electronic-structure prediction of superconducting transition temperature**; George Mason University, advised by D. Papaconstantopoulos; augmented plane wave band structure feeding an electron–phonon coupling estimate and a McMillan/Allen–Dynes transition temperature for intermetallic compounds. Undergraduate thesis.

electronic-structure calculation · augmented plane wave method · electron–phonon coupling · superconducting transition temperature of intermetallic compounds · computational materials screening

2015 **Dependent and directed type systems for multiphysics models**; Santa Fe Institute; proposal that multiphysics models be written in a dependent type language: classical fields are sections of fiber bundles and dependent types are fibrations, so model structure, including direction of coupling, can be carried in the type and exploited by the compiler.

multiscale and multiphysics modeling · dependent and directed type systems · compositional model building · category theory · scientific computing

2012 **Standing-wave mode structure of an RF discharge in a polyhedral cavity**; 2.45 GHz argon discharge in an evacuated aluminum polyhedron; the cavity geometry selects the supported resonant modes. Benchtop analogue for deployable conductive structures in ionospheric plasma.

RF plasma · standing-wave mode structure · cavity resonance · electromagnetics · deployable structures in ionospheric plasma

2012 **Flexible TiO_2 resistive-switching devices**; Mary Baldwin College; fabrication and bend-cycle characterization of memristive devices on compliant substrates, toward colocated storage and computation for body-worn sensing.

memristive devices · flexible TiO_2 resistive switching · thin-film fabrication · bend-cycle characterization · in-memory computing

FRAGMENTS

Structure recovered from the traces a language leaves behind.

2013 **Unsupervised grammar and lexicon induction from partially parallel text**; George Mason University; probabilistic lexicon induction over weakly parallel corpora, evaluated on contracted Grade 2 Braille as a test language with known ground truth; generalizes to incompletely deciphered scripts. Undergraduate thesis.

natural language processing · unsupervised grammar and probabilistic lexicon induction · partially parallel corpora · decipherment · contracted Braille · accessibility

2013 **Abstractive summarization of biomedical literature for clinical readers**; George Mason University; automated computational semantics aimed at reducing a paper to a triage-sufficient summary that keeps its evidence intact.

natural language processing · automated computational semantics · abstractive summarization · biomedical text · evidence fidelity

POSITIONS

2023–Present **CRC Postdoctoral Fellow**; Universität Münster; Arithmetic and Homotopy Theory Working Group, with Thomas Nikolaus and Christopher Deninger. Arithmetic patterns in homotopy theory and physics: geometric models of the Lubin–Tate action via moduli stacks of curves, and zeta functions in homotopy theory.

chromatic homotopy theory · moduli stacks of curves · arithmetic geometry

2015 **Visiting Researcher**; Santa Fe Institute of Complex Systems; Santa Fe, NM

2014–2016 **Thiel Fellow**; Thiel Foundation; Berkeley, CA; awarded at 17; medical and assistive technology, and independent mathematics study under Edward Frenkel

2013–2014 **Mouse Vocalization Specialist & Audio Processing Engineer**; Vium; Belmont, CA

2013 **Software Engineering Intern**; Cloudera; Palo Alto, CA; Hadoop · Hive · large-scale data processing

FELLOWSHIPS, AWARDS AND COMMUNITY

2022	**Gelfand Prize**, Northwestern University; for outstanding progress in research and extraordinary contributions to the research goals of the graduate community
2017	**NSF Graduate Research Fellowship**
2014	**Founding contributor, Topos House**; a still-active house in South San Francisco for scientific and artistic collaboration, especially between mathematicians and biologists
2013	**Outstanding Rising Senior Award** in Computational and Data Sciences, SPACS

EDUCATION

2017–2023	**Northwestern**; PhD in Mathematics; advised by Paul Goerss • *A Geometric Model of Higher K-theories at Height $h=p^{k-1}(p-1)$ via Families of Ramified Curves*
2015–2017	**UChicago**; Divisional Masters Program in Mathematics; advised by Peter May and Kazuya Kato
2012–2013	**George Mason University**; B.S. in Computational Physics (graduated winter 2013, at 16); advised by Dimitrios Papaconstantopoulos
2010–2012	**Mary Baldwin College**; Program for the Exceptionally Gifted: Mathematics, Computer Science, and Physics

METHODS

Inference	unsupervised learning and clustering · probabilistic lexicon induction · adaptive filtering and decoder analysis · policy search · evidence synthesis
Signals	real-time audio processing · neural signal decoding · computer vision · visual SLAM · sensor fusion · optical and immunoassay measurement
Modeling	electronic-structure calculation · multiscale and multiphysics modeling · type-theoretic model structure · large-scale data processing (Hadoop, Hive) · high-performance computing
Mathematics	homotopy theory · arithmetic geometry · moduli theory · category theory and type theory
Software	Python · NumPy · SciPy · scikit-learn · Git · Linux/Unix

WRITTEN WORK

SCIENCE: THESES, REPORTS AND WRITE-UPS

2015	***Simplifying Multiscale Modeling***; Santa Fe Institute of Complex Systems
2014	***A New Female–Female Mouse Vocalization Discovered via Unlabeled Machine Learning*** and ***On the Detection and Prevention of Aggression in Lab Mice via Quasi-Real Time Analysis***; internal write-ups, Vium
2013	***Contextual Machine Learning through the Analysis and Chunking of Partially Translated Grade 2 Braille***; undergraduate thesis, George Mason University
2013	***Predicting Superconductivity Transition Temperature via the Augmented Plane Wave Model***; undergraduate thesis, George Mason University
2013	**Nonprovisional patent application**; *Robotic Mobility Assistive Wheelchairs*; written, deliberately not filed, released under Creative Commons

MATHEMATICS: SELECTED PAPERS (FULL LIST AT RIN.IO/ABOUT)

Sept 2026	***On the Chromatic Splitting Conjecture at Coheight 1***; joint with T. Barthel, L. Mann, A. Senger, T. Schlank, J. Weinstein, X. Zhou; preprint
Sept 2025	***Moduli Stacks of G-Curves in Homotopy Theory at h = p−1***; arXiv:2509.23428; updated Sept 2026
July 2025	***Towards the p = 3 Kervaire Invariant Problem***; joint with E. Belmont; arXiv:2507.10157
2022	***Automorphisms of Abelian Varieties and Principal Polarizations***; joint with D. Lee; Rendiconti del Circolo Matematico di Palermo, Series 2, vol. 71, pp. 483–494; with a code base written for high-performance computing
2020	***Towards Directed Collapsibility***; joint with R. Belton, R. Brooks, S. Ebli, L. Fajstrup, B. T. Fasy, N. Sanderson, E. Vidaurre; Advances in Mathematical Sciences, vol. 21, pp. 255–271